MILITARY SPECIFICATION SHEET

ELECTRON TUBE, GAS

TYPE 869B

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Rectifier, half-wave, mercury

See figure 1

Mounting position: Vertical, base down

Weight: 1.5-pounds (680.4 grams) nominal

ABSOLUTE RATINGS:

Parameter:	Ef	epx	ib	Io	THg	F	tk
Unit:	Vac	kVac	a	Adc	°C	Hz	sec
Rectifier (1):							
Maximum:	5.25	10.0	10.0	2.5	60	150	---
Minimum:	4.75	---	---	---	30	---	60
Rectifier (2):							
Maximum:	5.25	15.0	15.0	5.0	40	150	---
Minimum:	4.75 (Note 1)	---	---	---	30	---	60
Rectifier (3):							
Maximum:	5.25	20.0	10.0	2.5	40	150	---
Minimum:	4.75	---	---	---	30	---	60
TEST CONDITIONS:	5.0	25.0	---	---	30-40	60	60

GENERAL:

Qualification - Required

Ⓑ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
							MIN	MAX	
	Quality conformance inspection, part 1								
1231	Peak emission by voltage drop	-	ib = 15a; THg = 30° to 40°C	0.65	II	etd	---	18.0	v
1301	Filament current	-		0.65	II	If	17	21	A
1353	Operation of rectifiers	2,3	Io = 2.5 Adc (min)	0.65	II	---	---	---	---
	Quality conformance inspection, part 3								
---	Service-life guarantee	-		---	---	t	1,000	---	hrs

NOTES:

1. For filament excitation 90° ± 30° out of phase with anode potential.
2. For operation test, one-half specified inverse voltage may be applied initially and then gradually increased to full voltage in approximately 1 minute.
3. This test is to be the first test performed at the conclusion of the holding period.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 80
DSA - ES

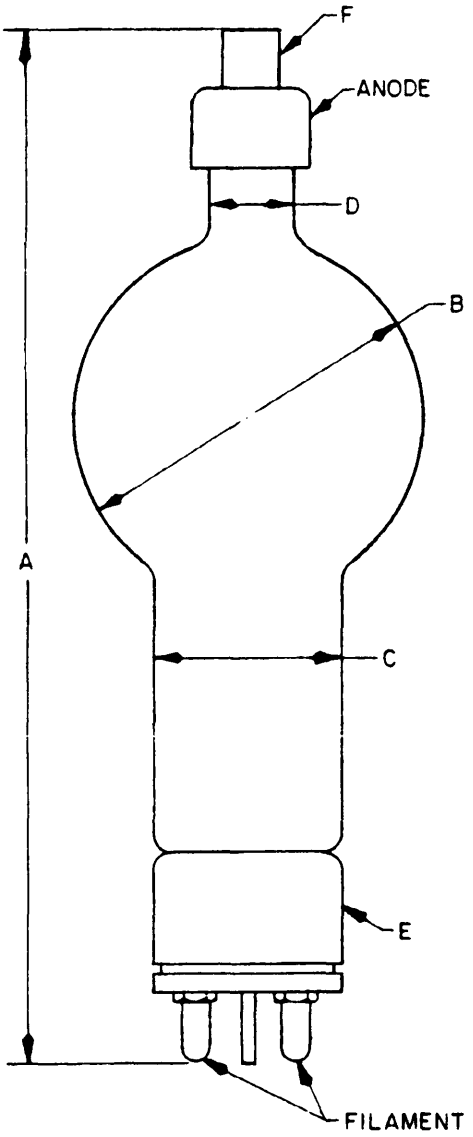
User activities:
Army - MU
Navy - AS, OS, MC, CG, SH
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-2921)

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	14.125 (358.78)	14.375 (365.13)
B	---	5.125 (130.18)
Quality conformance inspection, part 3 (see note)		
C	2.500 (63.50)	2.750 (69.85)
D	1.000 (25.40)	1.500 (38.10)
E	Base: A3-20 (EIA)	
F	Cap: C1-9 (EIA)	

NOTE:
Dimensions shall be checked during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

FIGURE 1. Outline drawing of electron tube type 869B.

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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OMB Approval
No. 22-R255

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   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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   ☐ YES ☐ NO *(If "Yes", in what way?)*

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